

Tracker Video Analysis & Modeling

Comprehensive User Manual & Physical Kinematics Reference Guide

1. System Overview & Core Architecture

Tracker is a computer-vision video analysis and physical modeling suite built for physics students, educators, and researchers. The workstation provides precision measurement tools to extract quantitative time-series data from real-world digital video recordings or synthetic physics simulations.

The system incorporates sub-pixel point digitization, interactive coordinate transformations, numerical finite-difference kinematics, energy analysis, mathematical regression curve fitting, and a full 3D stereo reconstruction engine with an interactive spatial orbit viewer.

2. Complete Interface Directory & Control Map

Every control, input, and instrument across the desktop interface is mapped below according to its primary functional zone:

A. Desktop Header & File Bar (Zone 1)

Control / ID	Element Type	Function & Scientific Purpose
Experiment Selector	Dropdown	Loads preset experiments (Projectile, 3D Helical, Pendulum, Free Fall, Incline).
Open Video (btn-upload-video)	Button (Upload)	Imports local MP4, WebM, or MOV video files for analysis.
Export CSV (btn-export-csv)	Button (Download)	Exports full kinematic table as standard comma-separated values (.csv).
Export Project (btn-export-trk)	Button (FolderOpen)	Serializes entire workspace (tracks, points, calibration, axes) into .trk project.
Help Guide (btn-help-guide)	Button (HelpCircle)	Opens the built-in quick guide modal with physics reminders.

B. Scientific Toolbar Instruments (Zone 2)

Toolbar Tool / ID	Icon & Type	Functional Behavior
Track Selector	Dropdown	Switches active particle (e.g. Mass A, Cart 1) and displays point count.
Track Manager (btn-track-manager)	Settings2	Configures mass (kg), point footprint (circle/diamond/square), and colors.
Clear Points (btn-clear-track-points)	Trash2	Clears digitized coordinates for active particle without deleting calibration.
Clip Cuts (btn-video-cuts)	Scissors	Opens timing dialog: Start/End Frame, Step Size, Frame Rate (FPS), t0.
Axes Toggle (btn-toggle-axes)	Compass	Toggles Cartesian coordinate frame (0, 0) and axis directions on canvas.
Axes Angle (axes-angle-quick-input)	Number Input	Rotates coordinate frame by theta degrees for inclined planes or tilted rigs.
Grid Toggle (btn-toggle-grid)	Grid	Overlays metric Cartesian grid lines over the video frame.
2D Calibration Stick	Ruler	Toggles blue 2-point reference ruler for pixel-to-meter scaling.
Calibration Settings	Sliders	Sets known physical length (m, cm, mm, ft) for the calibration stick.
3D Calibration (btn-3d-triangulation)	Sliders	Opens 3D Stereo Triangulation modal when in 3D Stereo mode.
Trails / Vectors / Loupe	Toggle Group	Toggles trajectory paths, velocity/accel arrows, and 3.5x magnifier.
Auto-step (btn-toggle-auto-advance)	FastForward	Automatically steps video forward by stepSize after marking a point.
2D / 3D Mode Switcher	Segmented	Switches between single-camera 2D and dual-camera 3D Stereo mode.
Layout Presets	Button Group	Switches view: Split, Graphs, Table, Video, Dual+3D, and 3D Orbit.

C. Transport Control Bar & Timeline Scrubber (Zone 5)

Control / ID	Icon / Type	Operational Description
Frame Input (direct-frame-input)	Number Field	Direct numeric jump to any target frame index.
Timeline Slider (video-frame-slider)	Range Scrubber	Continuous scrub bar displaying vertical tick marks for marked frames.
Time Readout	Monospace	Displays time $t = (\text{frame} - \text{startFrame}) / \text{FPS}$ to 3 decimal places.
Jump to Start (btn-first-frame)	SkipBack	Rewinds playback to the defined clip start frame.
Step Backward (btn-prev-frame)	ChevronLeft	Steps back by one stepSize frame increment.
Play / Pause (btn-play-pause)	Play/Pause	Toggles continuous playback. Keyboard shortcut: Spacebar.
Step Forward (btn-next-frame)	ChevronRight	Advances forward by one stepSize frame increment.
Jump to End (btn-last-frame)	SkipForward	Fast-forwards to the defined clip end frame.
Loop Toggle (btn-toggle-loop)	Repeat	Repeats playback automatically upon reaching the end frame.
Delete Mark (btn-delete-frame-point)	Trash	Removes the digitized point recorded at the current frame.
Playback Rate (playback-rate-select)	Dropdown	Selects playback speed (0.1x, 0.25x, 0.5x, 1.0x) for high-speed video.

3. Calibration Methods & Coordinate Systems

A. Standard 2D Calibration Stick Process

The 2D calibration stick establishes the linear scale factor S (pixels per meter) connecting video image space to real-world metric space. The pixel span between reference endpoints $A(u_A, v_A)$ and $B(u_B, v_B)$ is computed as:

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Pixel Span:   d_px = sqrt((u_B - u_A)^2 + (v_B - v_A)^2)
Scale Factor: S = d_px / L_known   [pixels / meter]
World X:      x = (u - u_origin) / S   (rotated by theta)
World Y:      y = (v - v_origin) / S   (rotated by theta)
```

- **Step 1:** Ensure camera optical axis was perpendicular to motion plane during capture.
- **Step 2:** Enable Calibration Stick on toolbar to display the blue reference bar.
- **Step 3:** Drag Point A and Point B end-caps across known object (e.g. 1.0 m stick).
- **Step 4:** Open Calibration Settings modal, enter known distance, choose units (m, cm, ft).
- **Step 5:** Click Apply Scale. All existing and future marks recalculate instantly.

B. 3D Stereo Calibration & Camera Matrix Configuration

To resolve true 3D coordinates (X, Y, Z) , Tracker incorporates two synchronized camera viewports with four selectable triangulation geometries:

- **Orthogonal Front-Side:** Camera 1 captures Front (X, Y) ; Camera 2 captures Side (Z, Y) . Both share the vertical Y axis.
- **Orthogonal Front-Top:** Camera 1 captures Front (X, Y) ; Camera 2 captures Overhead (X, Z) . Both share the horizontal X axis.
- **Convergent Stereo:** Dual angled cameras separated by baseline distance B and tilted by convergence angle β .
- **Direct Linear Transformation (DLT-11):** Maps 3D object space to camera pixels via 11 empirical camera calibration parameters $[L1..L11]$.

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DLT Forward Mapping (for each camera):
u = (L1*X + L2*Y + L3*Z + L4) / (L9*X + L10*Y + L11*Z + 1)
v = (L5*X + L6*Y + L7*Z + L8) / (L9*X + L10*Y + L11*Z + 1)
Triangulation solves least-squares ray intersection: [A]^T [A] [X, Y, Z]^T = [A]^T [B]
Reprojection Residual indicates spatial error (typically < 2.0 mm).
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4. Tracking Modes: Manual vs. Automated

Tracker offers two complementary tracking paradigms depending on video quality, target contrast, and occlusion conditions:

Feature Dimension	Manual Frame-by-Frame Digitizing	Automated Optical Tracker
Primary Use Case	Occluded, irregular, rotating, or low-contrast targets	High-contrast markers, solid balls, clean backgrounds
Sub-Pixel Inspection	3.5x Magnifier Loupe with reticle crosshair	Sub-pixel centroid template matching
Operator Interaction	Click target center of mass on each frame	Capture initial template, batch process sequence
Operational Speed	1 to 2 seconds per frame	Real-time multi-frame batch extraction
Error Handling	Operator drags mark to refine position anytime	Pauses on low cross-correlation confidence threshold

Manual Tracking Step-by-Step Workflow:

- **1. Enable Auto-step:** Ensure Auto-step is highlighted on the toolbar.
- **2. Activate Loupe:** Toggle Loupe to inspect the cursor area at 3.5x magnification with sub-pixel crosshairs.
- **3. Position & Click:** Click directly on the moving object centroid. World coordinates (x, y) record and the video advances automatically.
- **4. Coordinate Adjustment:** If a point is slightly off-center, scrub back and drag the point marker to the correct location.

Auto-Tracker Step-by-Step Workflow:

- **1. Select Start Frame:** Navigate to the frame where the target is distinct and clear.
- **2. Capture Template:** Click to place initial point. The tracker memorizes a target pixel patch around the centroid.
- **3. Search Window:** The optical correlation engine searches neighboring pixels on subsequent frames.
- **4. Supervised Review:** If target lighting shifts or cross-correlation drops, manually correct the point to resume.

5. Mathematical Kinematics & Energy Formulations

Derivatives are calculated using central finite differences to avoid phase lag inherent in forward or backward differences:

Internal Points (Central Difference):
vx(t_i) = [x(i+1) - x(i-1)] / [t(i+1) - t(i-1)]
vy(t_i) = [y(i+1) - y(i-1)] / [t(i+1) - t(i-1)]
vz(t_i) = [z(i+1) - z(i-1)] / [t(i+1) - t(i-1)]

Speed Magnitude: v = sqrt(vx^2 + vy^2 + vz^2)
Acceleration: ax(t_i) = [vx(i+1) - vx(i-1)] / [t(i+1) - t(i-1)]
Total Acceleration: a = sqrt(ax^2 + ay^2 + az^2)

Kinetic Energy: KE = 0.5 * m * v^2
Potential Energy: PE = m * g * y (where g = 9.80665 m/s^2)
Total Mechanical: E_total = KE + PE

6. Step-by-Step Laboratory Workflows

Tutorial 1: Standard 2D Projectile Motion & Gravity Extraction

- **Step 1: Video & Range:** Select "Projectile Motion" from the Experiment menu. Click Clip Cuts to confirm Start Frame, End Frame, and 30 FPS.
- **Step 2: Coordinate Setup:** Click Axes. Drag (0, 0) origin to ball launch position. Toggle Calib Stick, drag across 1.0 m reference, click Apply Scale.
- **Step 3: Point Digitizing:** Turn on Auto-step and Loupe. Click ball center on each frame until impact. Observe real-time trajectory trails and velocity vectors.
- **Step 4: Gravitational Fit:** Switch to Plot View. Set Y = y (Vertical Position), X = t (s). Select Parabolic fit ($y = A \cdot t^2 + B \cdot t + C$).
- **Step 5: Gravity Calculation:** From theoretical $y(t) = 0.5 \cdot g \cdot t^2 + v_0 \cdot t + y_0$, calculate $g_{exp} = 2 \cdot |A|$. Expected: $\sim 9.81 \text{ m/s}^2$ with $R^2 > 0.99$.

- **Step 6: Data Export:** Click CSV in header to export the dataset or Project to save a .trk session file.

Tutorial 2: Full 3D Stereo Triangulation Experiment

- **Step 1: Stereo Mode:** Switch toolbar mode from 2D to 3D Stereo. Switch Layout to Dual+3D.
- **Step 2: Camera Calibration:** Open 3D Calibration. Set method to Orthogonal Front-Side (90°). Verify scale factors and baseline distance.
- **Step 3: Dual Digitization:** On each frame, click the object in Camera 1 (Front), then click the object in Camera 2 (Side). The 3D engine computes (X, Y, Z).
- **Step 4: 3D Spatial Orbit:** In the 3D Orbit Viewport, drag to rotate yaw/pitch around the 3D trajectory. Use presets (Front, Top, Side, Isometric).
- **Step 5: Tabular Verification:** Open TableView. Verify calculated columns for z (m), vz (m/s), and az (m/s²).

7. Troubleshooting & Laboratory FAQ

- **Missing Initial Vectors:** Central difference derivatives require neighboring points (i-1 and i+1). Digitize at least 3 frames for velocity and 5 for acceleration.
- **Parabolic Coefficient g/2:** Remember that in $y = A*t^2 + B*t + C$, A represents $0.5*a$. You must multiply coefficient A by 2 to compute gravitational acceleration g.
- **Slow-Motion Video FPS:** If analyzing high-speed video (e.g. 240 fps), open Clip Cuts and set Frame Rate to 240. Otherwise, time steps will be calculated at standard 30 fps.
- **Reprojection Residuals:** A triangulation residual under 2.0 mm indicates accurate camera calibration and point marking. Higher values suggest optical miscalibration.